**System Design Documentation**
**Member Registration & Login System**

# Requirement Analysis

## System Domain

The Member Registration & Login System is a subsystem of the Springfield Library management platform.
It allows users to create accounts and log in using a library card number and PIN.
The system stores user data in Firebase Firestore and provides a simple authenticated home page.

## System Objectives

1. Provide an online form for new library member to register easily.
2. Validate user information before creating an account.
3. Generate a unique 8-digit Library Card Number for the users to use for login after they have successfully registered into the system.
4. Allow existing members to log in using Library Card Number + 4-digit PIN.
5. Store and retrieve user information reliably using Firebase (database).
6. Display a welcome message for user after successful login
7. User can logout and view dashboard

## component

| Component | Description |
|---|---|
| Register Page (Register.js) | Collects user information and creates a new member account. |
| Login Page (Login.js) | Authenticates users using card number + PIN. |
| Home Page (Home.js) | Displays a welcome message after the user successfully logs in, a logout option and a dashboard button to view the dashboard. |
| Firebase Member Service (members.js) | Handles registration, login, and database communication. |

| | |
|---|---|
| Firestore Database | Stores all member information after they register and access information for login. |

## Actors

1. Library member (user)
2. System
3. FirestoreDB

## functional requirements:

| **ID** | **Functional Requirement** | Description |
|---|---|---|
| **FR1** | User Registration | The system must allow users to create an account by entering full name, email address, phone number, date of birth and a 4-digit PIN (to be used for login) |
| **FR2** | Validate Input | The System must validate all inputs in the required fields correctly. |
| FR3 | Generate a library Card Number | The System must generate a unique 8-digit Library Card Number after the user registers. |
| FR4 | User Login | The System must authenticate the user with Library Card Number + PIN to login |
| FR5 | Welcome page display | The system must display a welcome window and an option to log out and view the dashboard after a successful login. |

## non-functional requirements:

| **ID** | **Functional Requirement** | Description |
|---|---|---|
| NFR1 | Performance | • Pages (Register, Login, Home) must load in under 2 seconds. |

| | | |
|---|---|---|
| | | • Login authentication must be completed in under 3 seconds.<br>• Registration submission must finish in under 5 seconds. |
| NFR2 | Usability | • Error messages must appear within 0.5 seconds after invalid input.<br>• A new user must be able to complete registration or login on their first attempt during usability testing. |
| NFR3 | Security | • User PINs must be stored securely.<br>• User data is stored securely in Firestore or localStorage. |
| NFR4 | Reliability | • The system must maintain 99% availability using Firebase services.<br>• Registration must save 0 incomplete records in the database. |

## Use case and case scenario:

### UC-1: Register a new member

Trigger: When the user clicks submit on the registration form.
Actors: Library Member (User), System, Database

Precondition:

- The User is on the Registration Page

Main Flow:

1. User enters full name, email address, phone number, date of birth, and a unique 4-digit PIN.

2. System validates that all fields are correct
3. System validates phone and PIN formats.
4. User click on the submit button
5. System generates a unique 8-digit Library Card Number.
6. System sends data to the database.
7. Database stores the user's information securedly.
8. System displays "Registered Successfully" with the library card number.

Postcondition:

- A new member record is stored in the Firebase database.
- The user receives an 8-digit Library Card Number.

Alternate Flows:

- AF1: Missing Fields
  – System displays "All fields are required."
  – Registration stops.

**UC-2: User login**

Trigger: User clicks the Login button.
Actors: Library Member (User), System, Database

Precondition:

- User must have already created an account
- User's data is stored in firebase.

Main Flow:

1. User enters their Library Card Number and 4-digit PIN.
2. System validates the input format.
3. System sends an authentication request to Firestore.
4. Firestore returns the matching member record stored in the database.
5. System verifies the card number and PIN that the user provided for login.
6. System redirects the user to the Home page.

Postcondition:

- User is authenticated and logged into the library's system.

Alternate Flows:

- AF1: Missing Input values
  – System displays "Both fields are required."
- AF2: Invalid PIN Format
  – System displays "PIN must be exactly 4 digits."
- AF3: Incorrect library card number
  – System displays "Invalid card number"

**UC-3: User logout**

Trigger: User clicks the Logout button on the Home Page.
Actors: Library Member (User), System

Precondition:

- User must be logged in to the system.

Main Flow:

1. User clicks Logout.
2. System clears user data from localStorage.
3. System redirects the user to the Login page.

Postcondition:

- User returns to the login page.

**UC-4: User view dashboard**

Trigger: User clicks the view dashboard on the Home Page.
Actors: Library Member (User), System

Precondition:

- User must be logged in to the system.

Main Flow:

4. User clicks view dashboard

5. System redirects the user to the dashboard page (another team’s subproject).

Postcondition:

- User views dashboard

## 1. Use Case Diagram

This Use Case Diagram visually represents how different actors interact with the Member Registration & Login System.

**Actors:**

1. Library Member: The primary end-user of the system. This actor performs all actions related to registration, authentication, and logging out.
2. System
3. Firestore Database: An external system responsible for persisting and retrieving user data.

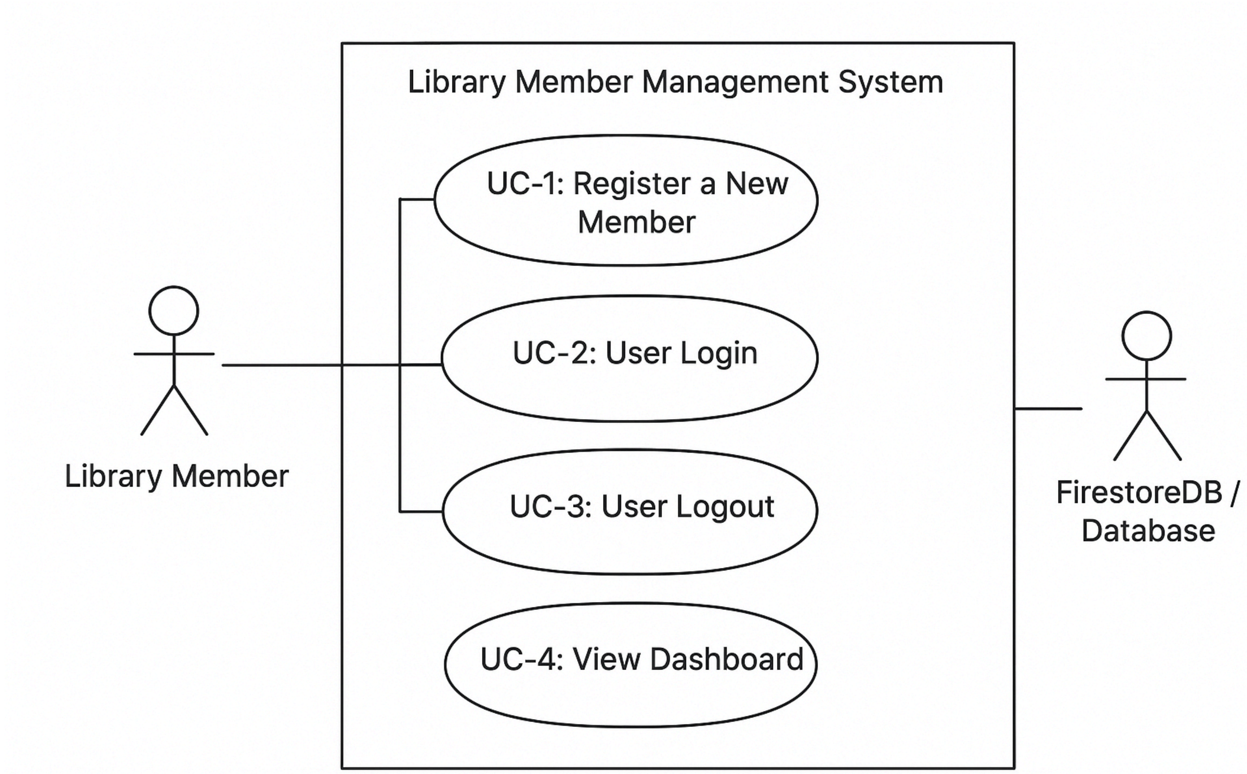
**Use Cases:**

1. Register New Member**:** Allows a new library user to create an account by providing personal details.
Initiated by the Library Member
The system saves details to Firestore

2. Login: Enables existing members to authenticate using their card number and PIN.
Validates credentials through Firestore queries

3. Logout: Safely ends the current session for the user.

4. User view dashboard: User clicks the view dashboard on the Home Page.



## 2. Class Diagram

### Key Classes

### 1.Member

Represents a registered user of the library.
Attributes include:

- cardNumber
- fullName
- Email
- Phone
- Dob

This class stores all the essential user information needed for authentication and identification.

### 2.RegisterPage / LoginPage / HomePage

These are UI-level classes:

- **RegisterPage:** Handles form submission for new users
- **LoginPage:** Manages login submission
- **HomePage:** Contains logout and welcome display functions

These classes separate the presentation layer from the logic layer.

### 3.MemberService

This is the core logic class.
It performs:

- Data validation
- Card number generation
- Member registration
- PIN verification
- Communication with the database

This ensures the UI pages remain clean and all business rules stay centralized.

### 4.FirestoreDB

Handles all interactions with Firestore.
Provides methods like:

- saveMember()
- getMember()

This abstraction allows the system to swap database technologies in the future without touching business logic.

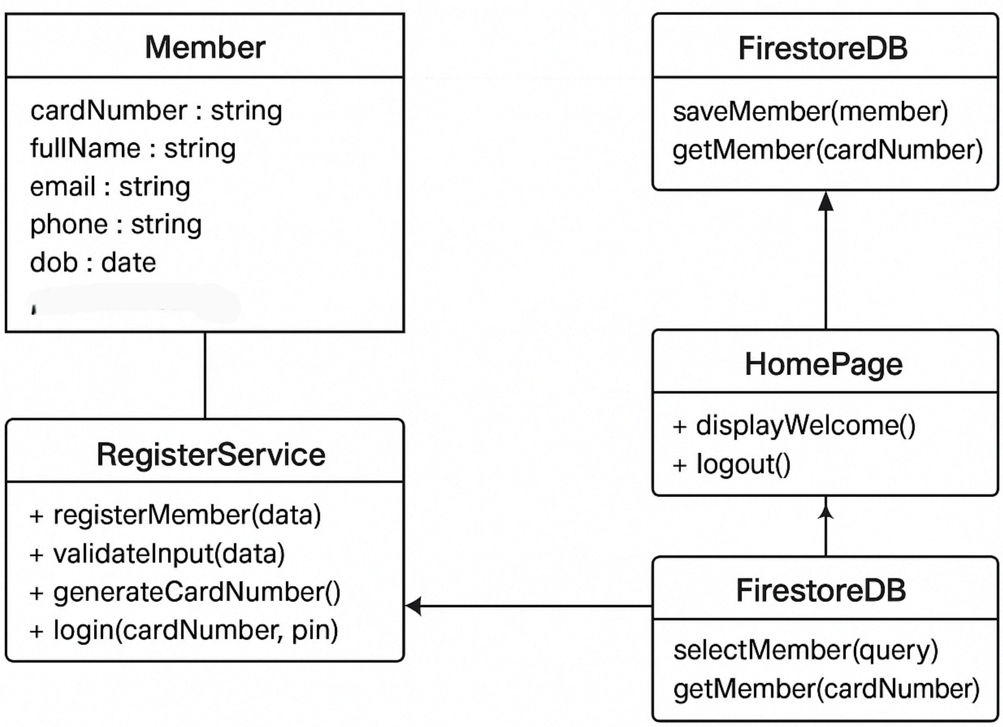
This class diagram follows a clean separation of concerns:

- UI
- Business Logic
- Database

This improves:

- Maintainability
- Scalability
- Testability

Every class has a single clear responsibility, making the system easier to extend or modify.

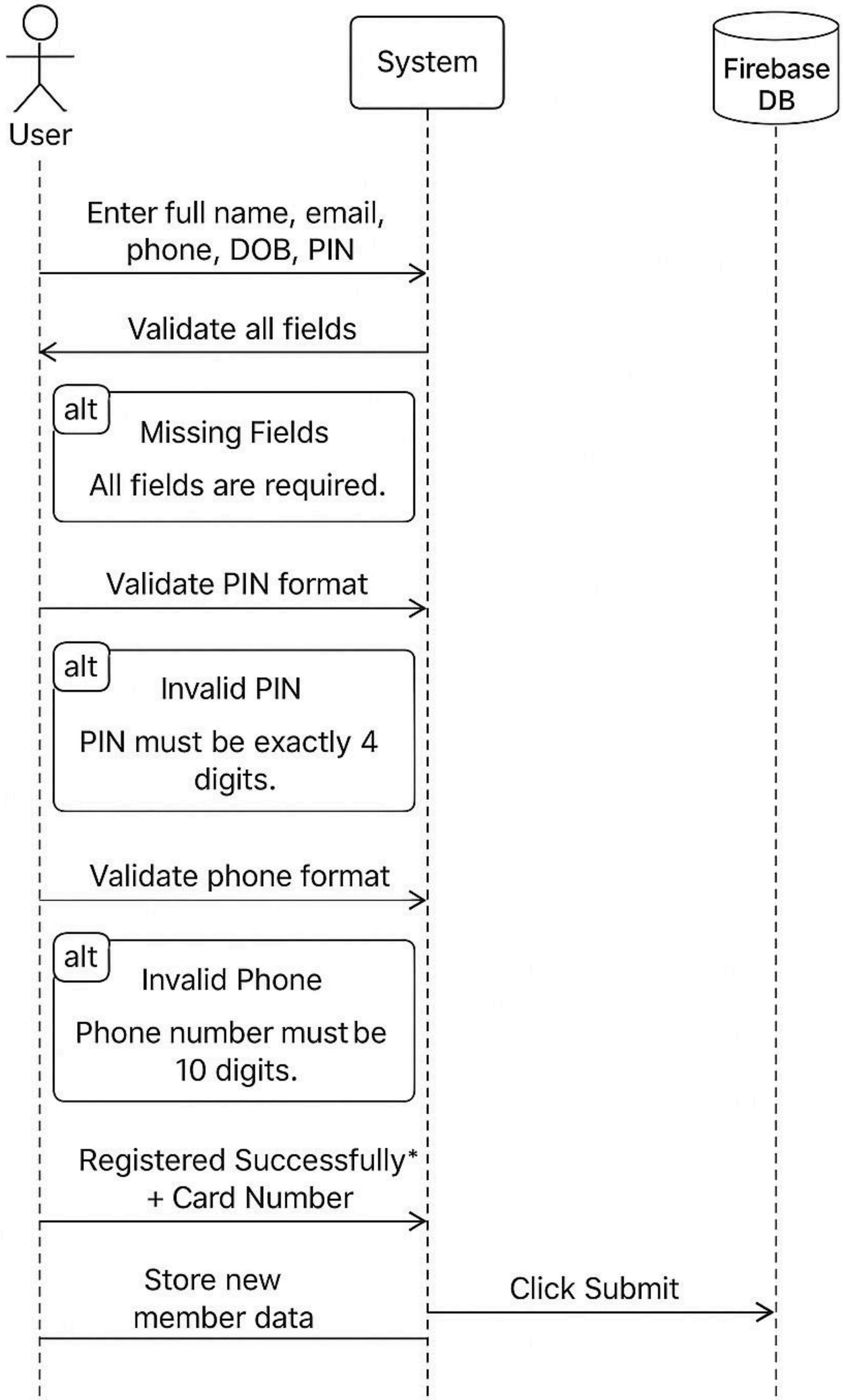


## 3.Sequence Diagrams

### UC-1: Register a New Member — Flow Summary

1. **User enters required details:** full name, email, phone number, date of birth, and PIN.
2. **System validates all fields.**
   - If any field is missing → System returns an error: *“All fields are required.”*
3. **System validates PIN format.**
   - If PIN is not exactly 4 digits → System returns *“PIN must be exactly 4 digits.”*
4. **System validates phone number format.**
   - If phone number is not 10 digits → System returns *“Phone number must be 10 digits.”*
5. Once all validations pass, the system confirms **Registration Successful** and generates a **Card Number**.
6. System sends the new member data to **Firebase Database** for storage.
7. User submits, and the registration process is completed.

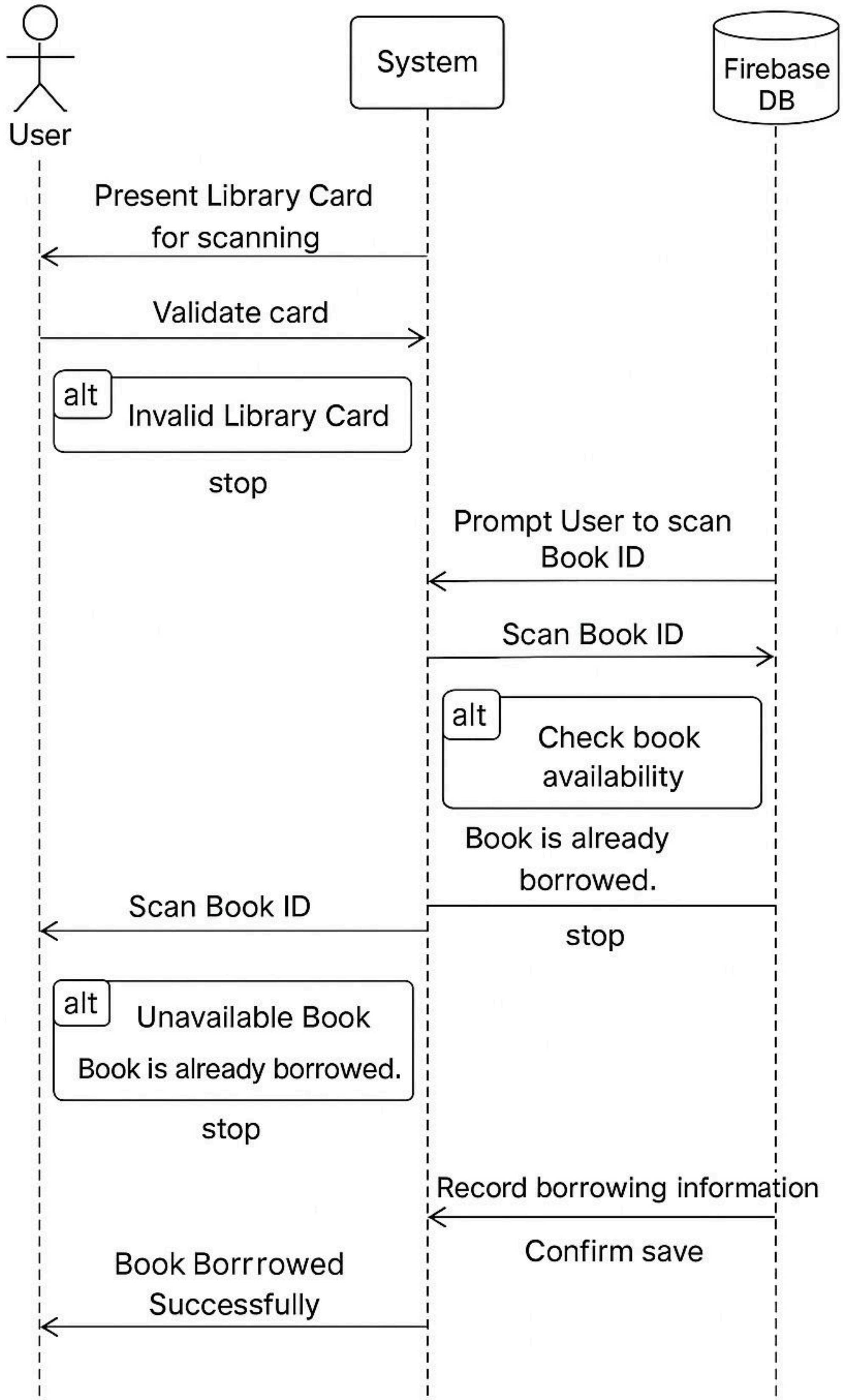
# UC-1: Register a new member



**UC-2: Borrow a Book — Flow Summary**

1. **User presents their library card** for scanning.
2. **System validates the card.**
    - If invalid → System returns *“Invalid Library Card”* and stops the process.
3. If the card is valid, the system **prompts the user to scan the Book ID**.
4. User scans the Book ID.
5. **System checks book availability.**
    - If the book is already borrowed → System returns *“Book is already borrowed.”* and stops the process.
6. If the book is available, the system records the borrowing information in **Firebase Database**.
7. Firebase confirms the save, and the system notifies the user: **“Book Borrowed Successfully.”**

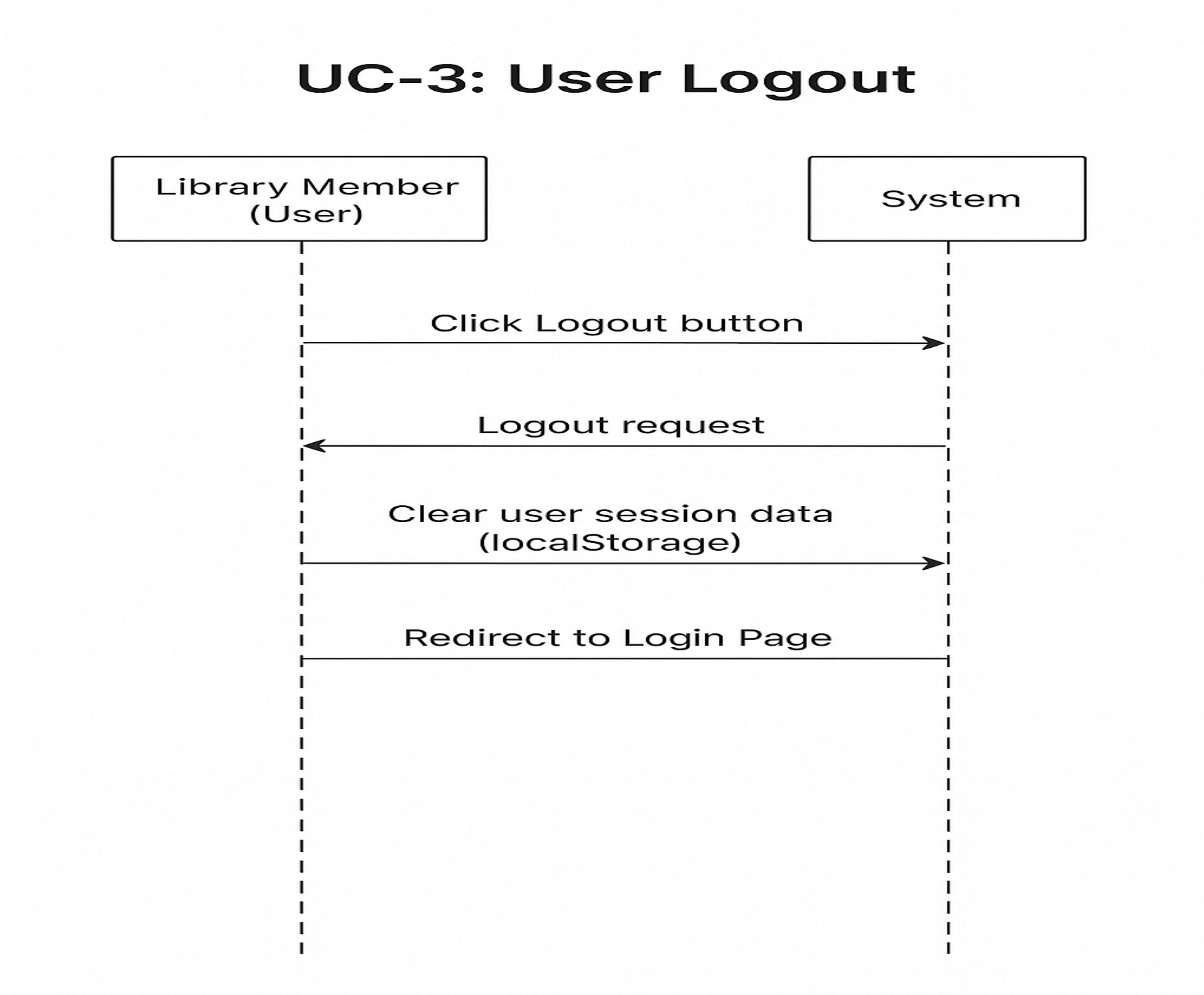
# UC-2: Borrow a book



**UC-3: Log Out of System — Flow Summary**

1. User clicks the Logout button on the Home Page.
2. System receives the Logout request.
3. System clears all stored user session data (localStorage).
4. System redirects the user to the Login Page.

Postcondition: User is successfully returned to the Login Page.




**3.4 Flow Summary for UC-4: User View Dashboard**

1. User initiates the action
   The Library Member clicks the “View Dashboard” button on the Home Page.

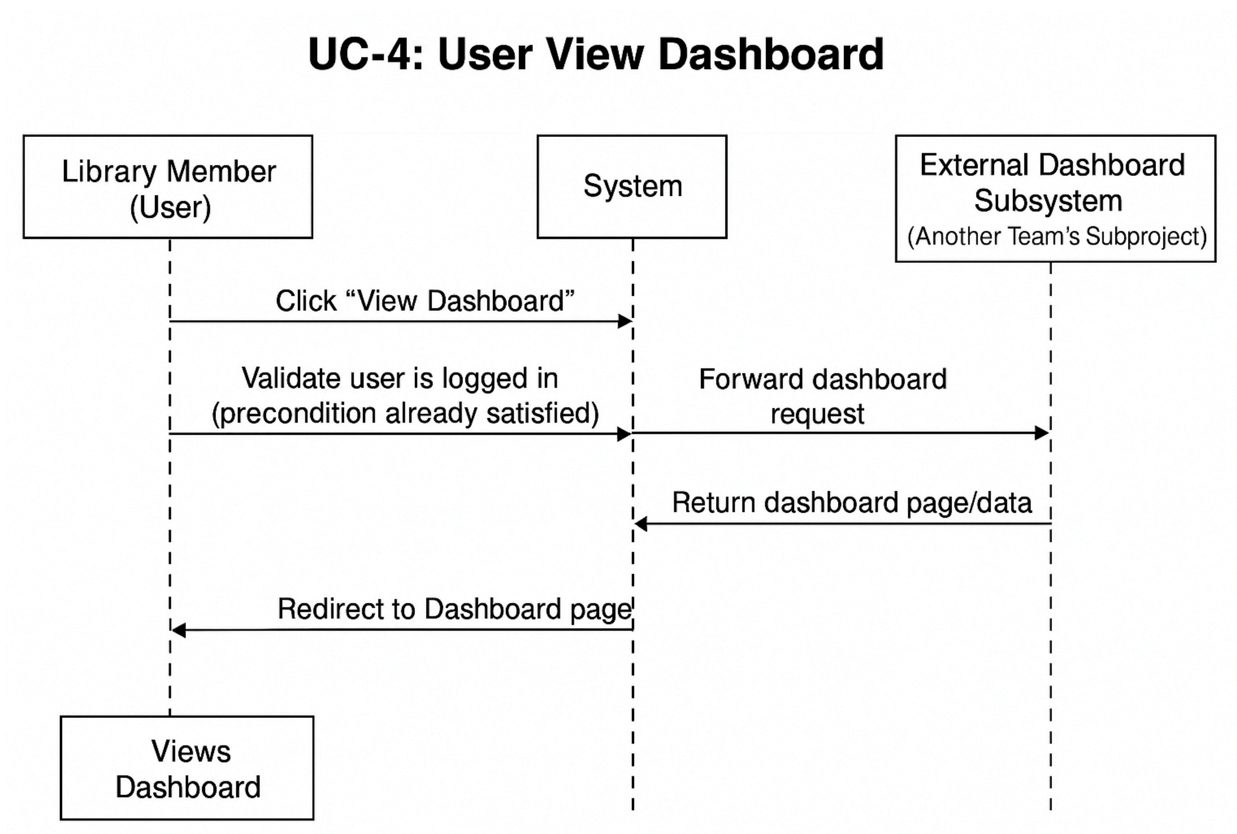
2. System receives the request
   The System validates that the user is logged in (precondition already satisfied).

3. System forwards the request
   The System redirects the request to another team's subproject, which hosts or generates the dashboard content.

4. Dashboard page returned
   The other team's subproject returns the dashboard page/data back to the System.

5. User is redirected
   The System redirects the Library Member to the Dashboard page.

Postcondition
The user successfully views their dashboard.




## 4. UI Wireframes Explanation

The wireframes describe the visual layout and fundamental UI structure of the application.
They are intentionally simple and clean for easy conversion into real HTML/CSS pages.

#### 4.1 Registration Page

**Features**

- Inputs for full name, email, phone, DOB, and PIN
- A clear "Register" button
- A link redirecting existing users to the Login Page

## Justification

- Follows standard onboarding flow
- All required profile data is collected at once
- PIN format is shown clearly to reduce user input errors
- Navigation is simple and accessible

**4.2 Login Page**

**Features**

- Card Number input
- PIN input
- Login button
- “Create Account” link for new users

**Justification**

- Minimal, focused design for quick access
- Clear field labeling prevents user mistakes
- Supports smooth flow between registration and login

**4.3 Home Page**

**Features**

- Personalized welcome message
- Display of the user's card number
- Logout button

**Justification**

- Confirms identity to the user

- Provides simple and safe exit via logout
- Represents a typical dashboard for a single-purpose system